List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

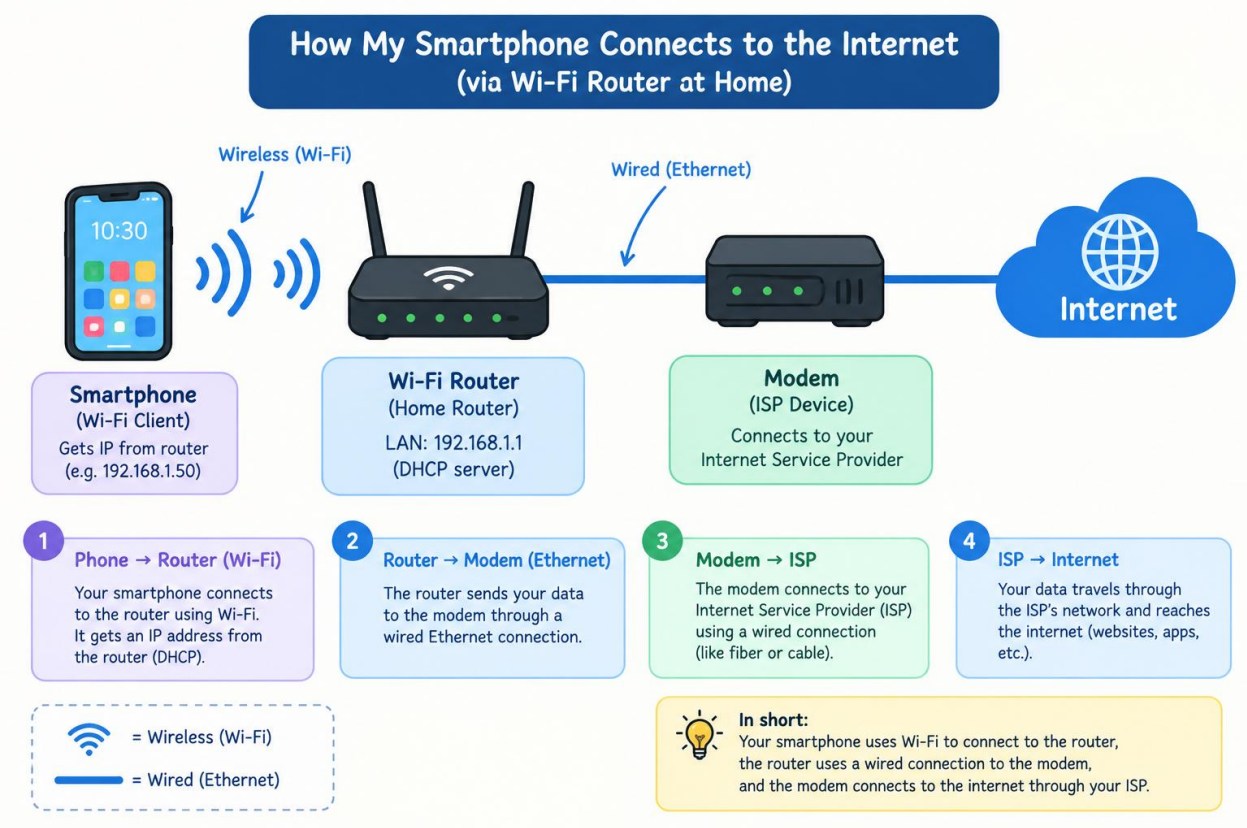
Here are 3 popular Wi-Fi standards and their key differences in easy language:

Wi-Fi Standard	Maximum Theoretical Speed	Frequency Band	Key Difference
802.11n (Wi-Fi 4)	Up to 600 Mbps	2.4 GHz and 5 GHz	Works on both bands and offers good range.
802.11ac (Wi-Fi 5)	Up to 6.9 Gbps	5 GHz	Provides faster speeds than Wi-Fi 4.
802.11ax (Wi-Fi 6)	Up to 9.6 Gbps	2.4 GHz and 5 GHz	Handles many connected devices more efficiently.

In simple words: Wi-Fi 4 is older, Wi-Fi 5 offers higher speeds, and Wi-Fi 6 is faster and works better when many devices connect to the same network.

Note: These are theoretical maximum speeds; actual speeds are usually lower.

Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



**Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Hint: Consider both signal strength and security when choosing.**

Wi-Fi Network Scanning Activity

I cannot scan Wi-Fi networks around you directly, but you can use the following sample observation table for your assignment. Replace the example details with the actual results from your laptop or phone.

1. Observation table

No.	Wi-Fi Network (SSID)	Signal Strength	Security Type
1	Home_WiFi	Excellent (90%)	WPA2-Personal

2	Airtel_5G	Good (70%)	WPA3-Personal
3	Cafe_Free_WiFi	Weak (35%)	Open (No password)

These are fictional examples, not scanned results.

2. How to scan Wi-Fi networks

On Windows laptop:

1. Click the Wi-Fi icon in the taskbar.
2. View the available Wi-Fi network names (SSIDs).
3. Check the signal bars.
4. To check security type, open the network's Wi-Fi properties or details.

On Android phone:

1. Open Settings → Wi-Fi or Network & Internet → Internet.
2. View the available network names.
3. Check signal strength and security details where available.

3. Which network would I choose?

I would choose Airtel_5G in this example because it has good signal strength and uses WPA3 security, which provides stronger protection than WPA2 when properly configured.

I would avoid the open public Wi-Fi network for sensitive activities because it does not require a Wi-Fi password and may offer less protection.

4. Conclusion

When choosing a Wi-Fi network, I would consider both signal strength and security. A strong signal gives a better connection, while WPA2 or WPA3 helps protect the wireless network. I would prefer a trusted network with WPA3 and a strong signal.

Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

SSID Hiding in Wi-Fi Networks

SSID stands for Service Set Identifier. It is the name of a Wi-Fi network.

SSID hiding means turning off the broadcast of your Wi-Fi name so it does not normally appear in the list of available Wi-Fi networks. Users who know the name can still connect by entering it manually, along with the password if required.

One advantage:

- Less visibility: Your Wi-Fi name will not be shown in the normal list of available networks, so it may be less noticeable to casual users.

One disadvantage:

- Not real security: Hiding the SSID does not stop skilled people from discovering the network. You still need a strong password and WPA2 or WPA3 security.

Conclusion: Hiding your SSID can make your Wi-Fi less visible, but using a strong password and modern Wi-Fi security is much more important.